

Economy, Ecology, and Social Welfare: A Win-Win Approach for Sustainable Tourism in Juneau

Summary

The development of tourism can lead to economic growth, but excessive development can cause ecological and social issues. This paper analyzes the situation of tourism in Juneau, Alaska, and formulates an optimization model, by which we draw the optimal development strategy, provide policy suggestions and applications in other scenarios.

For Problem 1, we devised an optimization strategy for Juneau's tourism industry, balancing economic profit and implicit costs. Using a **Multi-objective Nonlinear Programming Model based on Contingent Valuation Method (CVM) and Particle Swarm Optimization (PSO) Algorithm**, we converted implicit costs to dollars, forming a **total social utility** function. This function served as our objective, with **daily tourist numbers, government investment, project funding ratios, and tourist taxes** as decision variables. Our model yielded optimal values, leading to three policy suggestions: capping peak-season daily tourists at **13,993**, attracting **1,035** tourists daily in the off-season, and investing **\$92,310.46** daily in environmental and social infrastructure at a **54.33:45.67** ratio. Sensitivity analysis confirmed model stability, highlighting **peak-season tourist limits** as the key policy.

For problem 2, **Jiuzhaigou** in China is selected to verify the adaptability of the model. According to its characteristics like the number of tourists per day, relevant parameters are adjusted. It is concluded that the development of tourism in Jiuzhaigou area **does not need to directly limit the number of tourists in the peak season, but should take appropriate government investment as a more important policy**. Subsequently, the results are compared and analyzed based on the different characteristics between the two regions. Finally, to publicize less popular scenic spot, one effective way can be increasing the government's off-season expenditure to improve its attractiveness.

For problem 3, the optimal values of the decision variables in problem 1 are provided to the Tourist Council of Juneau as forecast data, and policy suggestions are provided from the perspective of **tourist quantity restriction, government investment in environmental and social construction, and government taxation on tourists**, among which the first policy is considered the most critical measure.

The model of this paper draws on the idea of CVM, enabling a more precise unification of previously ambiguous variables, and the robustness of the model results proves the rationality of this method, which is a highlight. Additionally, the model is comprehensive, adaptable, and has significant policy application value.

Keywords: Sustainable tourism; CVM; Multi-objective nonlinear programming